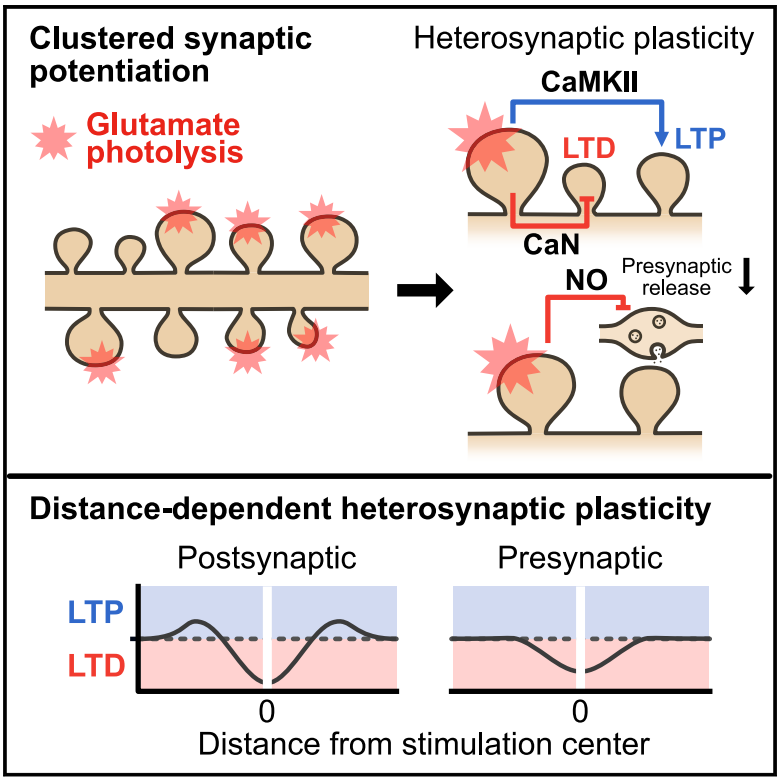
Cell Reports

Article

Heterosynaptic cross-talk of pre- and postsynaptic strengths along segments of dendrites

Graphical Abstract



Highlights

- Potentiation of spatially clustered synapses induces heterosynaptic plasticity
- Heterosynaptic plasticity is expressed at both pre- and postsynaptic terminals
- The polarity of neighboring pre- and postsynaptic plasticity is distance dependent
- Multiple signaling pathways differentially contribute to heterosynaptic plasticity

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In Brief

Tong et al. show that eliciting plasticity at groups of spatially clustered synapses triggers changes at neighboring synapses along the dendrite where spine plasticity is accompanied also by changes in presynaptic strength. The polarity of heterosynaptic plasticity is distance dependent, and CaMKII, calcineurin, and nitric oxide (NO) signaling play distinct roles.

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Heterosynaptic cross-talk of pre- and postsynaptic strengths along segments of dendrites

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SUMMARY

Dendrites are crucial for integrating incoming synaptic information. Individual dendritic branches are thought to constitute a signal processing unit, yet how neighboring synapses shape the boundaries of functional dendritic units is not well understood. Here, we address the cellular basis underlying the organization of the strengths of neighboring Schaffer collateral-CA1 synapses by optical quantal analysis and spine size measurements. Inducing potentiation at clusters of spines produces NMDA-receptor-dependent heterosynaptic plasticity. The direction of postsynaptic strength change shows distance dependency to the stimulated synapses where proximal synapses predominantly depress, whereas distal synapses potentiate; potentiation and depression are regulated by CaMKII and calcineurin, respectively. In contrast, heterosynaptic presynaptic plasticity is confined to weakening of presynaptic strength of nearby synapses, which requires CaMKII and the retrograde messenger nitric oxide. Our findings highlight the parallel engagement of multiple signaling pathways, each with characteristic spatial dynamics in shaping the local pattern of synaptic strengths.

INTRODUCTION

The maintenance of complex neural network dynamics imposes constraints on synaptic strength regulation. Cellular principles underlying neural network function, such as local dendritic integration or the minimization of metabolic cost, are thought to give rise to an optimal range of synaptic strength distributions, which, in turn, best support network function (Poirazi and Mel, 2001; Winnubst and Lohmann, 2012; Govindarajan et al., 2006; Harris et al., 2012). Indeed, synapses display several stereotypical organizational features of their strengths, including an apparent stable distribution across a synapse population that is skewed toward low magnitudes and a non-uniform spatial distribution (Buzsáki and Mizuseki, 2014; Barbour et al., 2007). How these synaptic patterns emerge from elementary molecular mechanisms is however not fully understood.

Previous studies have reported functional and anatomical similarities between neighboring synapses (Liang et al., 2018; McBride et al., 2008; Branco et al., 2008; Lee et al., 2019; Winnubst and Lohmann, 2012; Fu et al., 2012; Letellier et al., 2019; but see also Jia et al., 2010; Chen et al., 2011) that were mostly confined to dendritic branches. For instance, pre- and postsynaptic strengths were found to be more similar along short stretches of dendrites than across different branches of dendrites (Branco et al., 2008; Letellier et al., 2019). Similarly, synapses in close proximity were more likely to be synchronously active (Takahashi et al,. 2012; Winnubst et al., 2015; Kleindienst et al., 2011) and functionally correlated (Lee et al., 2019; Scholl et al., 2017). These synaptic patterns could be either explained by hardwired developmental mechanisms (Druckmann et al., 2014; Deguchi et al., 2011) or suggest additional activity-dependent mechanisms that facilitate coordination of synaptic strengths along short dendritic segments. In support of the contribution of activity-dependent mechanisms, pharmacological inhibition of NMDA receptor (NMDAR) activity abolished the spatial clustering of temporally correlated inputs (Winnubst et al., 2015; Kleindienst et al., 2011), and local dendritic activity was shown to adjust pre- and postsynaptic strengths (Branco et al., 2008; Letellier et al., 2019). Furthermore, functional synaptic clusters were more pronounced after environmental enrichment (Nikonenko et al., 2013) and prism rearing in barn owls (McBride et al., 2008). Alternatively, these local synaptic patterns may also arise as a consequence of local dendritic spiking (Poirazi et al., 2003; Branco and Häusser, 2010; Mel, 1992) or local homeostatic constraints on resources that promote the normalization of synaptic strengths in a homeostatic manner (Bourne and Harris, 2011; Sutton et al., 2006; Ju et al., 2004).

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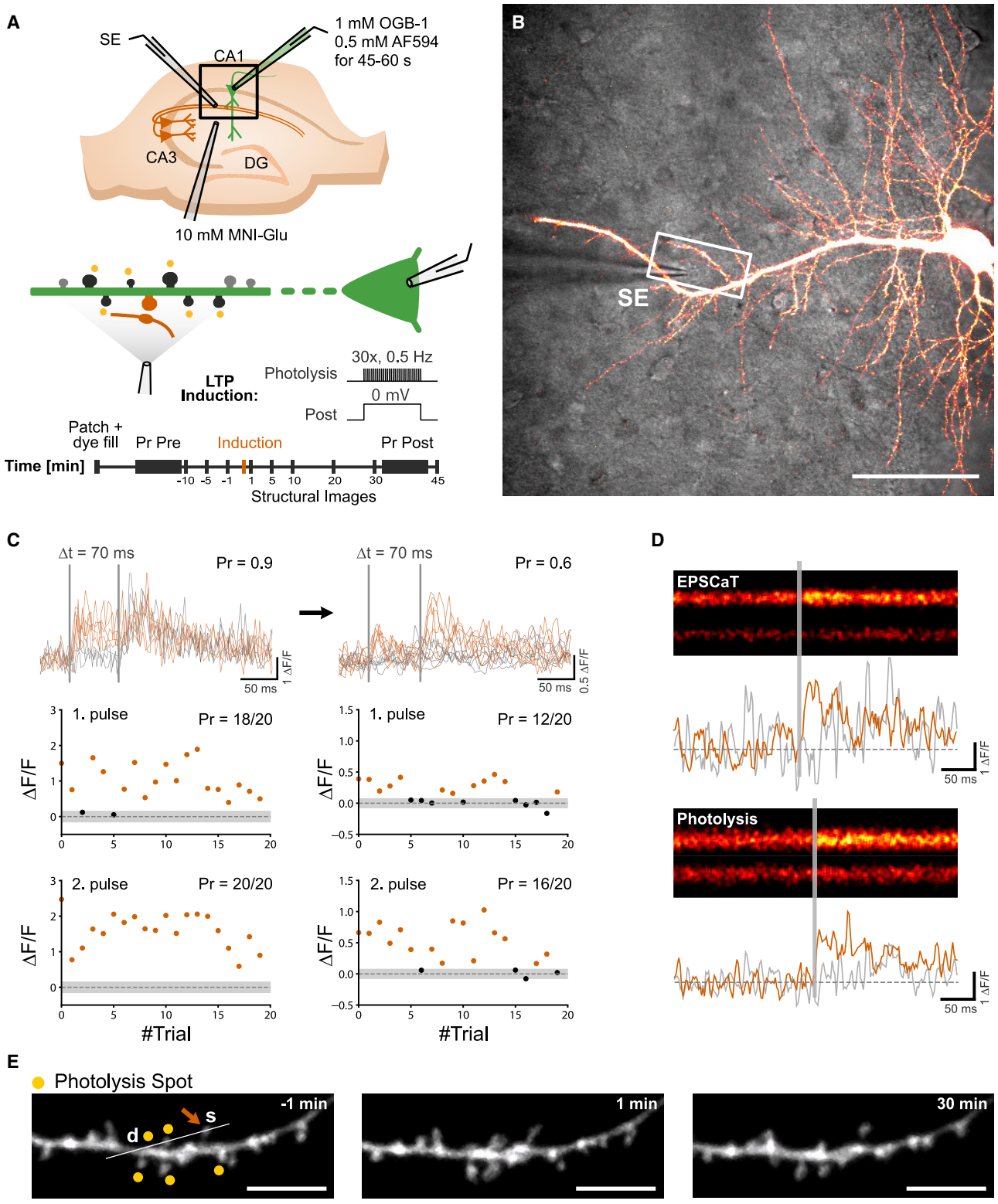


Figure 1. Experimental setup to study pre- and postsynaptic heterosynaptic plasticity

(A) Schematic of the experimental setup (top), cl-LTP induction protocol (middle), and timeline (bottom).

(B) Example neuron filled with 1 mM OGB-1 and 0.5 mM AF594 for 1 min. The boxed area is shown in (E). SE, stimulation electrode. Scale bar: 20 μm.

(legend continued on next page)

Collectively, these observations suggest that neighboring synapses interact with each other and that synaptic plasticity plays a role in implementing this coordination and possibly also competition between synapses. In agreement with such a view of intersynaptic interactions, a large body of evidence shows that activity-dependent changes of synaptic strength occur at both active (homosynaptic plasticity) and inactive synapses (heterosynaptic plasticity) (Royer and Paré, 2003; Schuman and Madison, 1994; Scanziani et al., 1996; Lynch et al., 1977). For instance, changes in synaptic strength at active synapses can either spread to adjacent synapses (Engert and Bonhoeffer, 1997; Wiegert and Oertner, 2013; Hayama et al., 2013; Letellier et al., 2019), induce a compensatory response at adjacent synapses (Bian et al., 2015; Oh et al., 2015; El-Boustani et al., 2018; Lee et al., 2013), or affect metaplastic properties to bias future modifications (Harvey and Svoboda, 2007; Harvey et al., 2008; Fonseca et al., 2004; Govindarajan et al., 2011). A better understanding of when, where, and how neighboring synapses interact will help provide insights into the emergence of higher level organizational principles of synapses, such as the minimal computational unit, and how its bound is set (Govindarajan et al., 2006).

Although a variety of phenomena and mechanisms have been reported for synaptic interactions, a common framework to explain the spatial coordination of synaptic strengths has been difficult to formulate (Kruijssen and Wierenga, 2019). For instance, it has been suggested that low-level organizational patterns might emerge from the unique spatiotemporal properties of signaling molecules that interact during synaptic plasticity (Nishiyama and Yasuda, 2015). However, the interplay of parallel plasticity pathways during synaptic plasticity has been difficult to observe and dissociate experimentally. In addition, even though it is generally accepted that synaptic plasticity occurs at both pre- and postsynaptic loci (Bliss and Collingridge, 1993) and heterosynaptic signaling between postsynaptic terminals along dendrites has been explored (Chater and Goda, 2020), little to nothing is known about heterosynaptic plasticity at the presynaptic terminal. This knowledge gap can mainly be attributed to the technical difficulties of measuring presynaptic strength, especially in more intact systems, such as brain slices, or *in vivo* (O'Rourke et al., 2012; Burette et al., 2015). However, investigating synaptic plasticity at both loci is particularly important because pre- and postsynaptic terminals might represent functionally distinct and independently regulated compartments (Padamsey et al., 2017; Costa et al., 2017; Xu et al., 2013; Sáez and Friedlander, 2009; Letellier et al., 2019) and are therefore likely to also differ in their spatial regulation with consequences on the property of information transfer (Costa et al., 2017; Tong et al., 2020).

Here, we have addressed the cellular basis underlying the organization of neighboring synaptic strengths at the Schaffer collateral inputs to the dendrites of CA1 pyramidal neurons. Our findings reveal a spread of plasticity along dendrite segments with a pattern distinct between the pre- and the postsynaptic sides, and our observations point to the engagement of independent signaling pathways that operate in parallel.

RESULTS

Local structural potentiation of groups of spines induces bi-directional postsynaptic heterosynaptic plasticity

We first investigated the synaptic cross-talk that might contribute to the formation of stable functional clusters. We hypothesized that the local coordination of synaptic strength is implemented by a tight balance of cooperation and competition between synapses. This predicts that experimentally imposing strong coordinated activity locally at a subset of synapses will induce a reorganization of synaptic strengths at all nearby synapses due to underlying competitive processes. We therefore emulated the activation of groups of spatially clustered synapses along segments of dendrites by using two-photon glutamate uncaging. Quasi-synchronous stimulation of groups of synapses was paired with cell-wide depolarization of the postsynaptic neuron to promote dendritic signaling events (Figure 1A). Postsynaptic strength was monitored by estimating the changes in spine volume, a parameter known to tightly correlate with the number of functional postsynaptic α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) receptors (Matsuzaki et al., 2001, 2004; Smith et al., 2003; Béïque et al., 2006). As a readout of presynaptic strength, we used optical quantal analysis to determine release probability (Pr) at selected synapses (Padamsey et al., 2019). Briefly, CA1 pyramidal neurons in organotypic hippocampal slices were loaded (45–60 s) by the patch pipette with the Ca^{2+} indicator OGB-1 (1 mM) and a fluorescent dye (AF594; 0.5 mM) to visualize fine dendrites. Dendritic segments of interest were locally stimulated to measure Pr of select synapses (Figure 1; Pr was estimated as the fraction of trials that resulted in successful release to the first pulse of pairs of stimulation pulses, as monitored by an excitatory postsynaptic Ca^{2+} transient [EPSCaT]; see STAR methods). For glutamate uncaging, 4-methoxy-7-nitroindolinyl (MNI)-caged glutamate (10 mM) was locally perfused, and the uncaging laser power was adjusted to match the magnitude and temporal dynamics of Ca^{2+} responses evoked by electrical stimulation (Figure 1D). To elicit long-term potentiation (LTP), the same neuron was repatched, and five to seven adjacent dendritic spines were

(C) Example of Pr measurements by optical quantal analysis before (left) and after (right) cl-LTP induction. Raw fluorescence recording traces (top) sorted into two groups showing either successful (red) or unsuccessful release (gray) to the first action potential. The plots below show the peak intensity of the response to the first (middle) and second pulse (bottom). The thick gray bar corresponds to two standard deviations of the baseline used as detection criteria for automated analysis. Pr was estimated as the fraction of trials that triggered successful release (top right corner). Facilitation to the second pulse of the pairwise stimuli is evident as an increase in Pr.

(D) Glutamate photolysis was adjusted to elicit Ca^{2+} transients similar in strength compared with EPSCaTs (laser power varied from 2.4–3.5 mW measured at the sample). The gray trace corresponds to the nearby dendrite.

(E) cl-LTP was induced by quasi-synchronous photolysis of glutamate at five to seven spines paired with postsynaptic depolarization, 30× at 0.5 Hz. An increase in spine size can be seen immediately after photolysis. For optical quantal analysis, line scans were taken through the spine of interest (s) and the dendritic branch (d). See STAR methods for details. Scale bar: 10 μm.

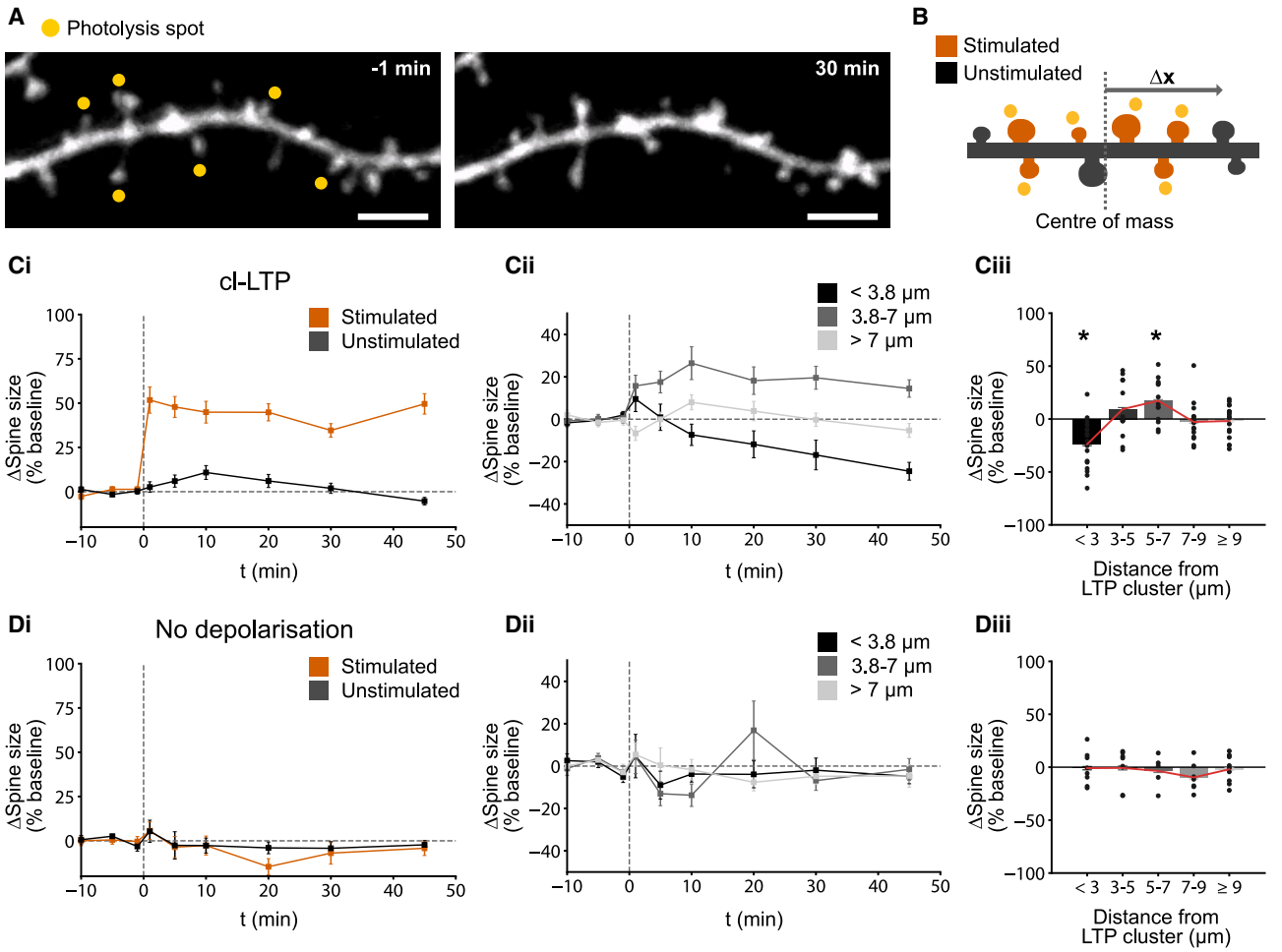


Figure 2. Induction of structural plasticity of groups of spines leads to heterosynaptic bi-directional changes in spine size

(A) Example of cl-LTP induction at six spines by glutamate photolysis. Spine structural changes were assessed for 30–45 min. Scale bar: 5 μm.

(B) The distance metric used for analysis of the spine cluster was defined as the distance along the dendritic branch to the center of mass of the stimulated cluster (Δx).

(Ci) Spine size changes of stimulated and unstimulated spines. cl-LTP was induced at t = 0 min. Stimulated spines showed robust structural LTP, whereas unstimulated spines enlarged transiently (n = 17). (Cii) Spine size changes of unstimulated spines, grouped by distance (see also Figure S4). Spines located close to the stimulated cluster decreased in size. More distal spines, however, increased in size. (Ciii) Persistent spine size change at 30–45 min depends on the distance to the stimulated cluster. Spines belonging to within 3 μm and 5 to 7 μm groups were significantly different from control experiments summarized in (Diii) ($p < 0.05$, Mann-Whitney U test). The polarity change associated with distance dependency of the mean spine size change is highlighted in red.

(D) Same as (C), but postsynaptic depolarization was prevented by holding the neuron at −70 mV during cl-LTP induction (n = 9). Neither stimulated nor unstimulated spines showed substantial structural changes. Data points are shown as mean ± SEM. See also Figures S1 and S2.

targeted for glutamate photolysis (30×; 0.5 Hz) paired with depolarization of the patch-clamped cell to 0 mV (Figure 1E). LTP was induced within 10 s of establishing the whole-cell patch mode, and the neuron was immediately patched off after induction. The re-patch led to a slight drop in the concentration of the fluorescent dye delivered during the first patch due to intracellular dialysis, which could confound the estimation of spine volume. We therefore normalized all spine fluorescence measurements to the average fluorescence intensity of the underlying dendritic branch (see STAR methods), which was assumed to remain largely structurally stable (Nimchinsky et al., 2004). We re-examined presynaptic strength after 30 min by repeating optical quantal analysis; spine structure was monitored for up to 45 min post-induction. We term LTP induced by this form of clustered stimulation as "cl-LTP."

Clustered stimulation led to structural enlargement of spine head volumes of a large fraction of stimulated spines, which was stable for at least 45 min (cl-LTP; Figure 2Ci, 59.5% ± 4.1% of spines with >20% increase in spine size at 30–45 min; see Figures S1 and S2 for spine Ca^{2+} changes induced by stimulation). The average increase in spine size after 30–45 min was $\Delta V_{30\text{-}45min}$ = 38.7% ± 4.5% of baseline (n = 17 experiments). Neighboring unstimulated spines also transiently increased in size shortly after cl-LTP induction (Figure 2Ci, ΔV_{10min} = 10.8% ± 4.0%) before slowly decaying back to baseline by 45 min ($\Delta V_{30\text{-}45min}$ = −1.1% ± 2.2%). When the postsynaptic cell was

held at −70 mV during clustered stimulation, stimulated spines did not increase in size, neither transiently nor over the long term (Figure 2Di; $\Delta V_{30\text{-}45min}$ = −4.9% ± 5.9%; n = 9). Without postsynaptic depolarization, neighboring unstimulated spines also remained unchanged (Figure 2Di; ΔV_{10min} = −2.8% ± 4.2%; $\Delta V_{30\text{-}45min}$ = −3.1% ± 2.9%). Local application of AP5 (500 μM; locally perfused) also abolished both homo- and heterosynaptic spine structural changes (Figure S3Ai).

Spatial proximity plays an important role in heterosynaptic plasticity because spatially delimited diffusion of signaling molecules between synapses is thought to mediate the local coordination of synaptic strengths (Nishiyama and Yasuda, 2015). We therefore grouped unstimulated spines based on their distance along the dendrite to the center of mass of the stimulated cluster (Figure 2B; see STAR methods). Unstimulated spines within 7 μm of the stimulated cluster showed significant bi-directional structural changes (Figure S4). The distance dependence of the bi-directional changes was most apparent when we compared spines within and spines beyond 3.8 μm of the stimulated cluster (Figure S4). We found that spines located within 3.8 μm showed a consistent decrease in size, which lasted for at least 45 min (Figure 2Cii; $\Delta V_{30\text{-}45min}$ = −23.0% ± 5.3%). In contrast, spines at a distance of 3.8–7 μm exhibited weak but significant potentiation, which lasted for the duration of the experiment ($\Delta V_{30\text{-}45min}$ = 18.1% ± 4.0%). Spines located further away showed a slow but transient increase in size, reaching a maximum at 10 min after stimulation and decaying back to baseline by the end of the experiment ($\Delta V_{30\text{-}45min}$ = −2.3% ± 2.9%). Figure 2Ciii summarizes the distance-dependent bi-directional regulation of postsynaptic strength, which is absent without postsynaptic depolarization (Figure 2Diii; cl-LTP versus no depolarization, $p < 0.05$ for spines located <3 μm and 5–7 μm; Mann-Whitney U test) and when NMDAR signaling was blocked with AP5 (Figure S3).

Local structural potentiation of groups of spines induces presynaptic heterosynaptic weakening

Next, we examined presynaptic changes following cl-LTP. Due to the comparatively low throughput nature of optical quantal analysis in which Pr measurement in general was feasible for only a single synapse per experiment, we focused mainly on synapses located close to the stimulated cluster (synapse located <4 μm to the stimulated cluster in 13/17 experiments). cl-LTP induction resulted in a decrease in Pr of such proximal synapses (Figure 3C; ΔPr = −0.27 ± 0.05, n = 13, <4 μm), whereas Pr of synapses located more distally did not change (Figure 3C; ΔPr = −0.02 ± 0.06, n = 4, >4 μm). Given that the activation of presynaptic NMDARs can lead to the expression of presynaptic long-term depression (LTD; Padamsey et al., 2017), the large amounts of glutamate released during photolysis could have resulted in local accumulation and spillover of glutamate to cause a decrease in Pr at neighboring unstimulated synapses. However, holding the neuron at −70 mV during cl-LTP induction abolished presynaptic changes (Figure 3D; no depolarization: ΔPr = −0.02 ± 0.06, n = 11, $p < 0.01$, Kruskal-Wallis H-test with post hoc Dunn's test). Thus, glutamate uncaging by itself was not sufficient to induce heterosynaptic LTD (hetLTD) of Pr, and concurrent postsynaptic depolarization was required. Notably, postsynaptic depolarization alone is known to influence Pr (Branco et al., 2008; Volgushev et al., 1997). Therefore, to control for this possibility, we repeated the experiment without glutamate photolysis by either locally perfusing artificial cerebrospinal fluid (aCSF) without caged glutamate or by omitting the photolysis pulse while applying postsynaptic depolarization. Under these conditions, presynaptic weakening was not observed (Figure 3D; no photolysis: ΔPr = −0.01 ± 0.03, n = 18, $p < 0.01$, Kruskal-Wallis H-test with post hoc Dunn's test). Presynaptic hetLTD was also abolished by the local application of AP5 during the induction of cl-LTP (Figure S3C). Thus, presynaptic hetLTD requires both postsynaptic depolarization and synaptic stimulation of nearby synapses.

The decrease in Pr at synapses neighboring the stimulated spines was correlated with the initial Pr, which we did not observe for control conditions (Figure 3F). This dependence of the extent hetLTD of Pr on the initial Pr, which in itself was highly heterogeneous (mean initial Pr = 0.58, SD = 0.21), could explain the variance of the data for the change in Pr as a function of distance. Specifically, the magnitude of presynaptic changes was not strongly related to the distance to the stimulated cluster (Figure 3G). Instead, the observation of presynaptic hetLTD was limited to synapses located within ~4 μm of the stimulated cluster, unlike the magnitude of spine size changes that were correlated with the distance to the stimulated cluster for up to ~7 μm. Moreover, despite exhibiting some trend, the magnitudes of pre- and postsynaptic changes were not significantly correlated (Pearson r = 0.487, n = 13, p = 0.09; Figure 3H).

Altogether, strong stimulation of groups of synapses led to a restructuring of synaptic strengths along the dendrite. The postsynaptic strength of unstimulated synapses changed bi-directionally, which was dependent on the distance to the stimulated synapses. In contrast, the presynaptic terminal weakened only at synapses close to the stimulated cluster. The different patterns of heterosynaptic pre- and postsynaptic changes and the lack of correlation between pre- and postsynaptic changes at individual synapses suggest the parallel engagement of independent signaling pathways.

Presynaptic hetLTD requires CaMKII activation, whereas postsynaptic changes are driven by parallel signaling pathways involving calcineurin and CaMKII

Next, we sought to further confirm the suggested involvement of distinct signaling pathways by gaining insights into the underlying molecular pathways involved in the heterosynaptic cross-talk. The spatial reorganization of synaptic strengths might reflect the biochemical computations driven by the interaction of signaling pathways (Nishiyama and Yasuda, 2015). Strong evidence suggests that heterosynaptic postsynaptic weakening requires the activation and diffusion of the phosphatase calcineurin (Oh et al., 2015; Hayama et al., 2013). In contrast, the activation of the kinase CaMKII and its downstream targets including the small GTPases was reported to facilitate LTP at nearby synapses (Harvey and Svoboda, 2007; Harvey et al., 2008; Hedrick et al., 2016). Therefore, calcineurin and CaMKII are potential candidates for plasticity factors mediating the depression and potentiation, respectively. Our observation of a bi-directional regulation of postsynaptic strength could be explained by the orchestrated spatial diffusion of these two plasticity factors, with CaMKII causing

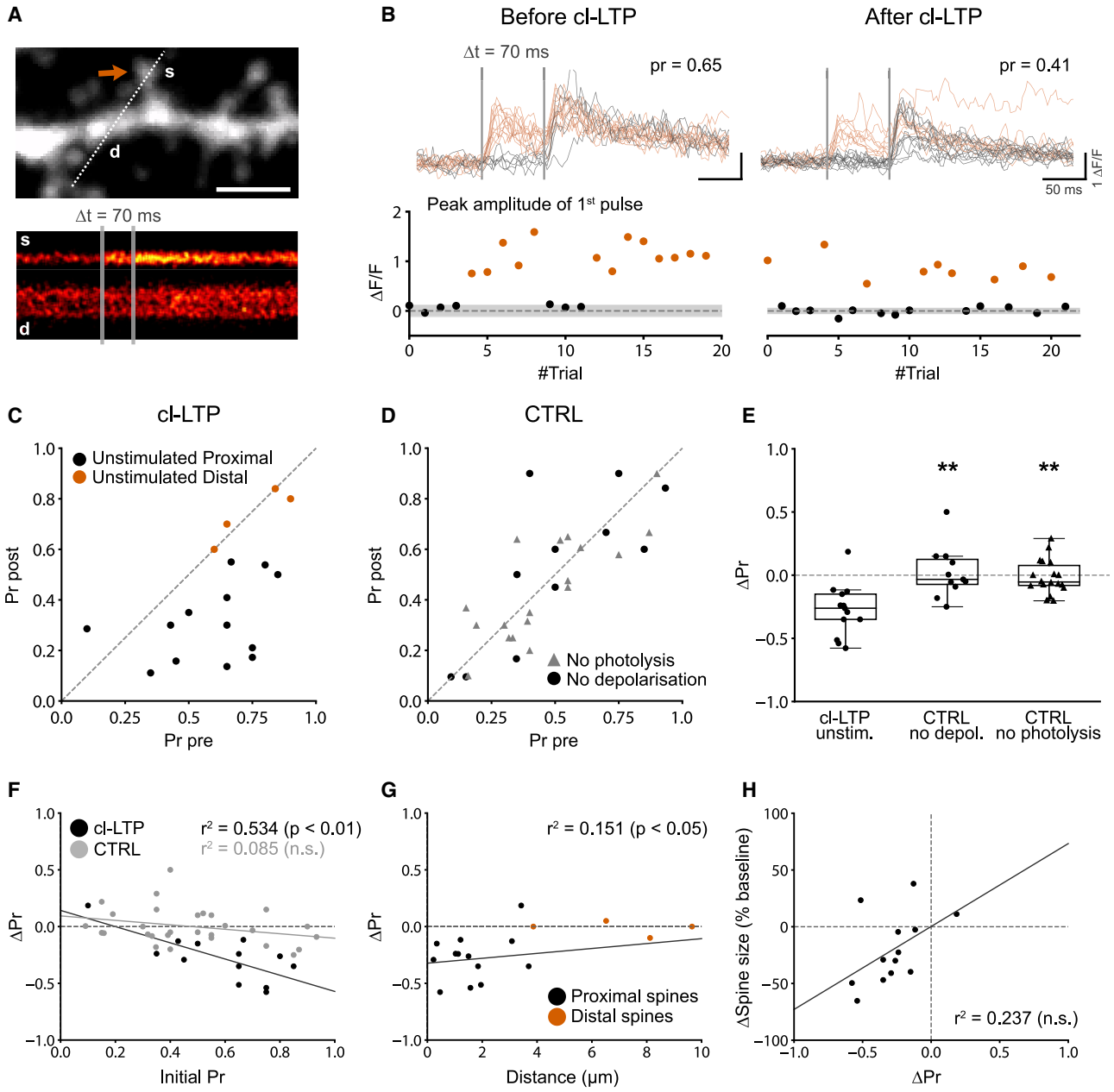


Figure 3. Induction of structural plasticity of groups of spines leads to heterosynaptic presynaptic weakening

(A) Pr was measured using optical quantal analysis. Top: Line scans were taken through the spine of interest (red arrow, s) and the dendrite (d) while the synapse was stimulated. Scale bar: 2 μm. Bottom: Glutamate release was detected as Ca^{2+} transients in the spine head (ESPCaTs).

(B) Top: Raw Ca^{2+} transients show all-or-none response profile to repeated stimulation, before (left) and 30 min after (right) the induction of cl-LTP. Vertical gray lines indicate time of stimulation. Responses to the first stimulation pulse were used to estimate Pr. Bottom: Peak amplitude of the EPSCaT of the first pulse is shown as function of trial number (inter-trial interval = 15 s). The gray bar represents two standard deviations from the baseline. An increase in release failures is observed 30 min after cl-LTP induction (right panel).

(C) Pr 30 min after cl-LTP induction (post) versus baseline Pr (pre). Pr decreased for synapses located close to the stimulated cluster (<4 μm; unstimulated proximal) but not in distal synapses (>4 μm; unstimulated distal).

(D) No Pr change was observed when omitting glutamate photolysis (no photolysis, triangles) or postsynaptic depolarization (no depolarization, circles).

(E) Summary of Pr changes (cl-LTP unstim, n = 13, all synapses within 4 μm of the stimulated cluster; CTRL no depol., n = 11; CTRL no photolysis, n = 18). Significance is shown for comparisons with label cl-LTP unstim ($p < 0.01$, Kruskal-Wallis H-test with post hoc Dunn's test).

(F) The change in Pr as function of the initial Pr.

(G) Pr changes relative to distance to the stimulated cluster.

(H) Correlation between pre- and postsynaptic heterosynaptic changes.

potentiation and calcineurin causing depression. We thus predicted that (1) selective inhibition of calcineurin should abolish the postsynaptic weakening of proximal synapses without affecting the potentiation at more distal synapses and (2) inhibition of CaMKII should lead to the opposite effect. To test these predictions, we repeated the experiment under the pharmacological blockade of either calcineurin or CaMKII.

The vehicle control recapitulated our previous results (Figure 2) with the exception of a slight shift in the spatial bounds that best explained the bi-directional heterosynaptic changes (Figure 4A; < 2.8 μm, $\Delta V_{30\text{-}45min}$ = −15.5% ± 4.7%; 2.8–6.5 μm, $\Delta V_{30\text{-}45min}$ = 11.4% ± 5.7%; >6.5 μm, 5.9% ± 3.1%, n = 19; see also Figure S4), which could have been attributed to culture variability. In the presence of bath-applied 10 μM KN62, an inhibitor of CaMKII, the cl-LTP of stimulated spines was blocked (Figure 4B; $\Delta V_{30\text{-}45min}$ = 6.4% ± 9.2%, n = 10, $p < 0.01$), consistent with previous reports (Matsuzaki et al., 2004). In the presence of KN62, unstimulated spines located close to the stimulated spines decreased in size, as observed above. However, similarly to the stimulated spines, the enlargement of spines located more distally was completely abolished (Figure 4Bii; <2.8 μm, $\Delta V_{30\text{-}45min}$ = −10.6% ± 8.2%; 2.8–6.5 μm, $\Delta V_{30\text{-}45min}$ = −6.9% ± 5.6%; >6.5 μm, $\Delta V_{30\text{-}45min}$ = −3.1% ± 3.2%). In contrast, inhibition of calcineurin by bath application of 2 μM FK506 completely abolished heterosynaptic postsynaptic weakening of proximal spines (Figure 4C). Instead, upon blocking calcineurin activity, we observed potentiation of both proximal and distal spines (<2.8 μm, $\Delta V_{30\text{-}45min}$ = 24.7% ± 8.8%; 2.8–6.5 μm, $\Delta V_{30\text{-}45min}$ = 17.7% ± 8.3%; >6.5 μm, $\Delta V_{30\text{-}45min}$ = 6.6% ± 4.9%, n = 8). FK506 did not affect the magnitude of cl-LTP at stimulated spines ($\Delta V_{30\text{-}45min}$ = 40.4% ± 4.3%). Data summary is shown in Figure 4D. Inhibition of CaMKII but not calcineurin also abolished presynaptic heterosynaptic weakening (Figure 4E).

Altogether, these experiments demonstrate a clear pharmacological dissociation of heterosynaptic postsynaptic potentiation and depression. Consistent with our hypothesis, our observations suggest that postsynaptic heterosynaptic plasticity is regulated by at least two opposing factors, which depend on calcineurin and CaMKII activity. Moreover, selective block of presynaptic but not postsynaptic weakening by KN62 also points to a dissociation between pre- and postsynaptic plasticity. Notably, upon inhibiting calcineurin, presynaptic weakening was accompanied by postsynaptic potentiation. Presynaptic hetLTD is therefore not likely caused by a matching between pre- and postsynaptic strengths (see e.g., Tang et al., 2016; Meyer et al., 2014; Bayazitov et al., 2007; Kay et al., 2011; Tokuoka and Goda, 2008) but is subject to regulation by an independent signaling pathway that acts in parallel to the postsynaptic regulatory mechanism.

NO is the retrograde messenger for presynaptic hetLTD

The heterosynaptic presynaptic weakening in our experimental paradigm necessitated some form of retrograde signaling, as the effects of stimulation targeted to the postsynaptic spines were prevented in the absence of depolarization of the postsynaptic neuron. The findings so far suggested that the retrograde messenger was likely produced downstream of NMDAR activation, involved CaMKII signaling either upstream or downstream, and was expected to rapidly diffuse over short distances from the stimulated synapse cluster. Any one of a range of known candidate molecules fulfills these requirements. In particular, strong local dendritic depolarization can trigger the production and release of nitric oxide (NO) (Padamsey et al., 2017), a highly diffusible signaling molecule. Furthermore, NO signaling is known to interact with NMDAR and CaMKII signaling (Hardingham et al., 2013). NO is also implicated in (homosynaptic) presynaptic LTP (Padamsey et al., 2017) and LTD (Stanton et al., 2003). We therefore hypothesized that strong local activity triggers the production and release of NO, which, due to its gaseous nature, diffuses over a confined distance to neighboring presynaptic terminals to elicit presynaptic weakening. We tested this hypothesis by applying the NO synthase inhibitor L-NAME.

Bath application of 100 μM L-NAME completely abolished presynaptic hetLTD (Figures 5B and 5D; ΔPr = −0.02 ± 0.06, n = 8, $p < 0.01$). In contrast, L-NAME did not impair cl-LTP induction and expression at stimulated spines ($\Delta V_{30\text{-}45min}$ = 53.6% ± 9.5%, n = 8), the transient increase in spine size immediately after induction (ΔV_{10min} = 11.7% ± 1.4%), or the distance-dependent bi-directional heterosynaptic regulation of spine size (Figure 5A). In fact, the fraction of stimulated spines expressing stable structural LTP was significantly larger when NO production was inhibited (77.4% ± 4.5% of spines with $\Delta V_{30\text{-}45min}$ >20%, $p < 0.05$ versus vehicle).

We next sought to further confirm the suggested role of NO in a more physiological experimental paradigm. To do so, we devised an experimental scheme by using acute brain slices that closely followed our optical setup in slice cultures. We assessed the plasticity of Schaffer-collateral-CA1 synapses within relatively close spatial proximity of stimulated synapses and yet at a distance at which basal responses remained independent (Figure 6; see also STAR Methods). Along individual stretches of dendrites, which were identified by an intracellular fluorescent dye (AF488) introduced to the patched CA1 neuron, we positioned two stimulation electrodes (SEs) 10–40 μm apart to increase the probability of stimulating sets of synapses located close to each other. We used a cross-facilitation test to ensure the stimulation of independent Schaffer-collateral pathways. We induced LTP in one pathway by pairing presynaptic stimulation with strong postsynaptic depolarization by current injection, which elicited bursts of action potentials 60 times at 5 Hz (Padamsey et al., 2017). We then examined changes in the unstimulated pathway.

Induction of LTP led to an increase in the excitatory postsynaptic potential (EPSP) slope of the stimulated pathway (Figure 6B; ΔEPSP = 94.9% ± 13.1%, n = 7) and a decrease in the paired-pulse ratio (PPR) of EPSP slopes (Figure 6C; ΔPPR = −0.203 ± 0.032), which was stable for at least 30 min, consistent with previous findings (Padamsey et al., 2017). This result supported a likely presynaptic expression locus. Conversely, the unstimulated pathway showed mild depression of the EPSP slope with an increase in PPR (ΔEPSP = −19.1% ± 15.4%, ΔPPR = 0.307 ± 0.092). The heterosynaptic PPR changes were strongly correlated with the fractional change of the EPSP slope (Figure S5; r = −0.842, $p < 0.05$), which suggested that electrically induced heterosynaptic plasticity was expressed as a decrease in Pr, a finding that was in agreement with heterosynaptic presynaptic weakening that accompanied

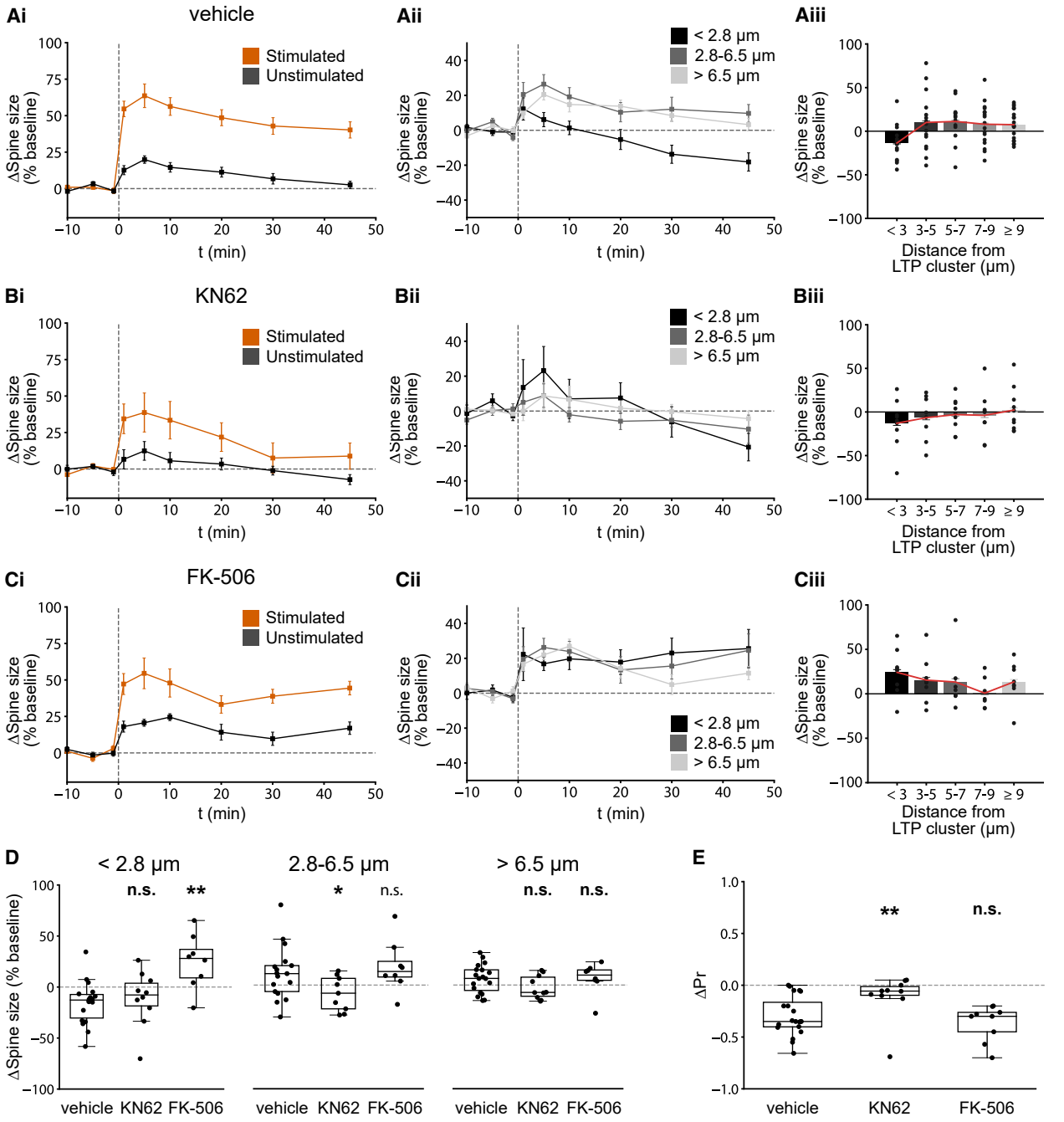


Figure 4. CaMKII is required for postsynaptic heterosynaptic LTP, whereas calcineurin is required for postsynaptic heterosynaptic LTD

(Ai) Spine size changes of stimulated (orange) and unstimulated (gray) spines in vehicle control group. Spine enlargement was robust and persistent at stimulated spines ($\Delta V_{30\text{-}45min}$ = 42.5% ± 5.1%) and transient at unstimulated spines (ΔV_{10min} = 14.6% ± 3.3%) (n = 19). (Aii) Spine size changes of unstimulated spines, grouped by distance. (Aiii) Spine size changes of unstimulated spine, grouped in 2-µm bins.

(Bi–Biii) Effects of bath-applied CaMKII inhibitor KN62 (10 µM) on spine size changes. The expression of homo- and heterosynaptic LTP but not LTD was abolished (n = 7).

(Ci–Ciii) Effects of bath-applied calcineurin inhibitor FK-506 (2 µM). Heterosynaptic shrinkage of proximal spines was completely abolished, and spine enlargement was observed instead (n = 8). Homosynaptic LTP remained unchanged ($\Delta V_{30\text{-}45min}$ = 40.4% ± 4.3%, n = 8). Error bars represent SEM.

(D) Comparison of heterosynaptic spine size changes for proximal (<2.8 µm), distal (2.8–6.5 µm), and more distant spines (>6.5 µm). Statistical comparison includes L-NAME experiments shown in Figure 5 for the multiple comparison.

(E) Comparison of heterosynaptic Pr changes for spines located within 4 µm of the stimulated spine cluster (Kruskal-Wallis H-test with post hoc Dunn's test). Significance indicates comparison with vehicle control group. Data points are shown as mean ± SEM. See also Figures S3 and S4.

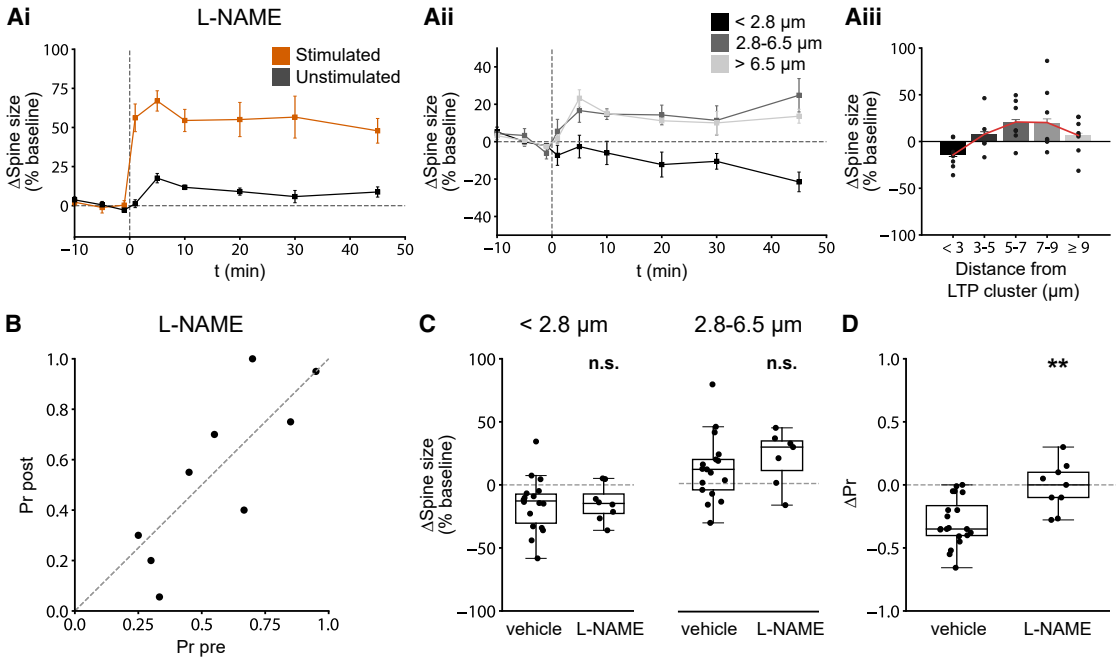


Figure 5. NO is the retrograde messenger for heterosynaptic presynaptic weakening

Spine size changes in the presence of bath-applied NO synthase inhibitor L-NAME (100 μM). (Ai) Spine size changes of stimulated and unstimulated spines (n = 8). (Aii) Spine size changes of unstimulated spines, grouped by distance (<2.8 μm: $\Delta V_{30\text{-}45min} = -14.3\% \pm 4.7\%$; 2.8–6.5 μm: $\Delta V_{30\text{-}45min} = 20.3\% \pm 6.9\%$; >6.5 μm: 11.7% ± 4.0%). (Aiii) Spine size changes of unstimulated spines, grouped in 2-μm bins. Inhibition of NO production by L-NAME did not affect spine size changes of either stimulated or unstimulated spines.

(B) Pr before and 30 min after the induction cl-LTP.

(C) Summary of spine size changes.

(D) Summary of presynaptic changes (n = 8, $p < 0.01$). Statistical significance was tested using Kruskal-Wallis H-test and post hoc Dunn's test, and comparisons were made with the vehicle control group. Data points are shown as mean ± SEM.

cl-LTP in slice cultures. In contrast, PPR changes of the homosynaptic pathway were not correlated with the EPSP slope change ($r = 0.373$, not significant), which suggested the occurrence of a postsynaptic change that was concomitant to the expression of presynaptic plasticity. Next, we inhibited NO synthase by bath application of 100 μM L-NAME. As observed previously (Padamsey et al., 2017), this abolished the homosynaptic presynaptic LTP (Figures 6D and 6E; $\Delta PPR = -0.029 \pm 0.052$, $n = 7$, $p < 0.05$). Furthermore, the heterosynaptic increase in PPR was also blocked and a stable increase of the EPSP slope was unmasked ($\Delta EPSP = 56.0\% \pm 12.3\%$, $\Delta PPR = 0.020 \pm 0.036$, $p < 0.05$). Altogether, our data suggest that NO signaling orchestrates both potentiation and depression of presynaptic strength in a context-dependent manner, that is, potentiation of release at stimulated synapses and depression of release at neighboring non-stimulated synapses.

DISCUSSION

Studies of heterosynaptic plasticity to date have mostly focused on local structural plasticity of spines along dendritic branches (Oh et al., 2015; Hayama et al., 2013; Wiegert and Oertner, 2013; Govindarajan et al., 2011; El-Boustani et al., 2018; Bian et al., 2015; ; Harvey and Svoboda, 2007; but see also Letellier et al., 2019). Both compensatory and non-compensatory forms of heterosynaptic changes relative to the direction of plasticity induced at the stimulated spines have been reported, whereas the underlying basis for when, where, and what form of heterosynaptic plasticity is expressed have remained unclear. Here, we found that heterosynaptic spine changes that likely represented both compensatory and non-compensatory processes could be induced at individual synapses by the same cl-LTP stimulus although with different spatial profiles along the dendritic segment. HetLTD was expressed at spines close to the stimulated synapses, whereas heterosynaptic LTP was observed at spines further away, which altogether resulted in a spatial profile that resembled a Mexican hat wavelet. What might constitute the emergence of this spatial profile? In light of previous findings, an attractive explanation for our observations could involve the differential properties of signaling pathways triggered by calcineurin and CaMKII and how they influence the neighboring spines. Calcineurin, which promotes structural depression of spines (Oh et al., 2015), is itself reported to diffuse into the dendrite and to reach neighboring spines over short distances (Fujii et al., 2013). Furthermore, small GTPases such as H-Ras, Rac, or Rho, which are downstream effectors of the potentiating factor CaMKII, also diffuse along the dendrite and can reach substantial distances (Harvey and Svoboda, 2007; Harvey et al., 2008; Hedrick et al., 2016) to mediate heterosynaptic metaplastic changes that manifest as a facilitation to subsequent

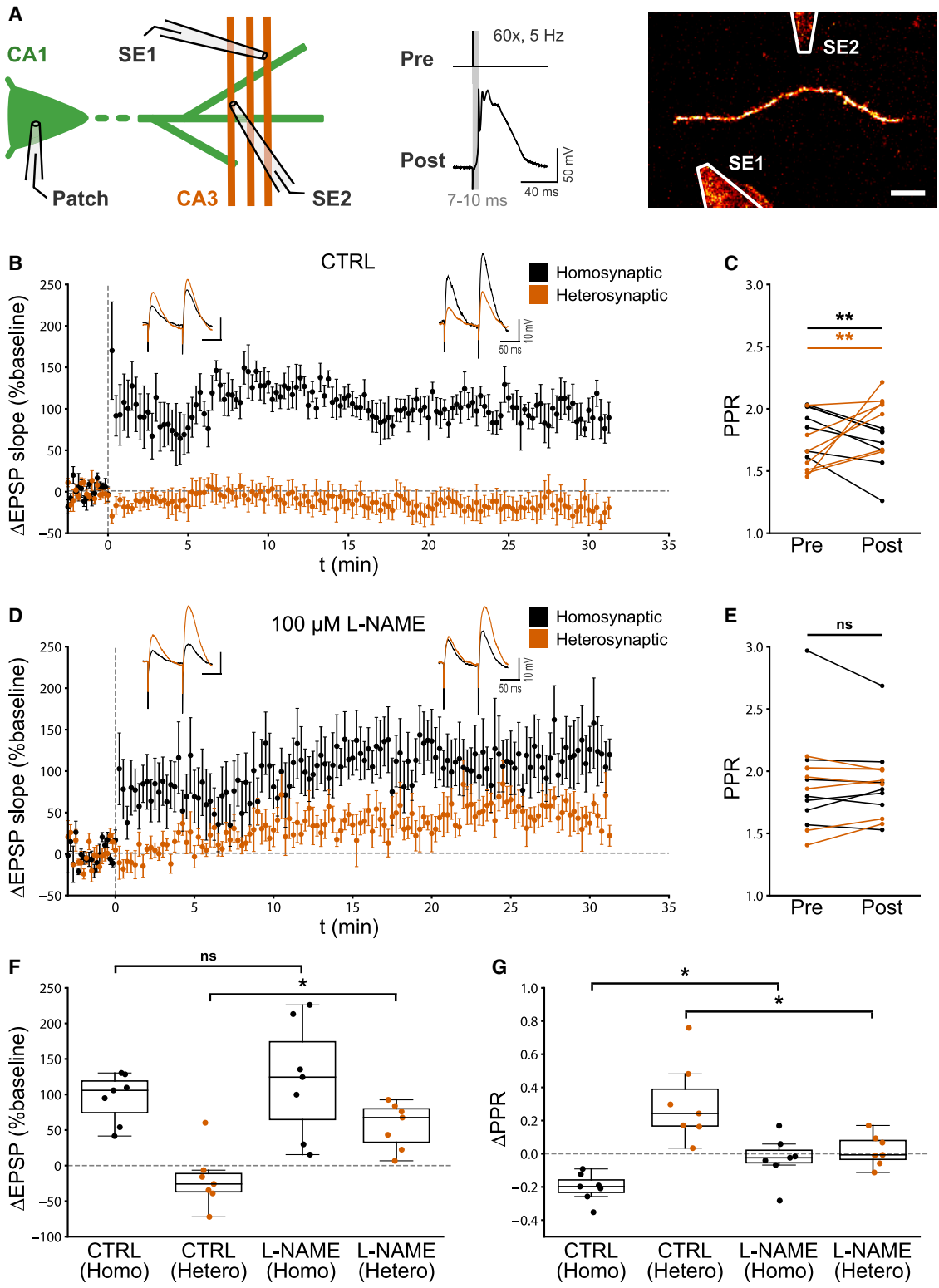


Figure 6. Heterosynaptic LTD in acute hippocampal slices is expressed presynaptically and requires NO signaling

(A) Experimental scheme. CA1 pyramidal neurons in acute hippocampal slices were filled by patching with an electrode containing 200 μM AF488 for 5 min. Two tungsten stimulation electrodes (SE1/SE2) were placed along a segment of dendrite, 10–40 μm apart. Spike-timing-dependent LTP was induced in one pathway by pairing presynaptic stimulation with postsynaptic current injection that elicited complex spikes (60× at 5 Hz). Scale bar: 10 μm.

(legend continued on next page)

induction of potentiation (Harvey et al., 2008; Hedrick et al., 2016). Therefore, in one possibility, differential diffusion along the dendrite of calcineurin and active small GTPases could in effect produce the Mexican hat wavelet profile of spine structural plasticity that we have observed. Such a scenario stipulates that the effective diffusion of calcineurin must be weaker than that of CaMKII-dependent signaling events, and this possibility remains to be tested.

The overall spatial pattern of heterosynaptic spine plasticity that we found was mostly consistent with previous reports of heterosynaptic plasticity apart from some notable differences, such as the presence of the Mexican hat profile discussed above. The heterosynaptic spine shrinkage that we observed was in accord with the study by Oh et al. (2015), in which the sequential potentiation of spines in close spatial proximity produced shrinkage of unstimulated spines within ~3.4 μm of the stimulated cluster in a process requiring the activation of calcineurin but not CaMKII. However, unlike our study, Oh et al. (2015) did not report an enlargement of spines located further away. Furthermore, several studies of single spine LTP showed heterosynaptic facilitation to subsequent induction of LTP (Harvey and Svoboda, 2007; Harvey et al., 2008; Hedrick et al., 2016). These studies, however, did not observe a shrinkage of nearby spines, which is in contrast to our study and the study by Oh et al. (2015). What might be the cause of these differences in heterosynaptic spine changes accompanying LTP induction at target spines? Previously, CaMKII activation was shown to display a highly supralinear response to stimulation frequency and number (Fujii et al., 2013). In contrast, calcineurin activation was sublinear and required a longer integration time window (Fujii et al., 2013) from multiple sources (Oh et al., 2015), suggesting that calcineurin activation might be strongly thresholded (Oh et al., 2015). Based on these findings, the qualitative differences in heterosynaptic spine dynamics might be explained not only by the possible contribution of the differential diffusion properties of signals associated with activation of calcineurin and CaMKII at stimulated spines (see above) but also by the differential activation of calcineurin and CaMKII at stimulated and neighboring spines. In other words, different stimulation used to trigger LTP would produce distinct spatiotemporal pattern of calcium increases along the dendritic segment (Figures S1 and S2), and this could in turn promote distinct activation patterns of calcineurin and CaMKII along the dendrite to bias the plasticity outcome at individual spines. Accordingly, the heterosynaptic spine shrinkage in the absence of potentiation of distal spines was triggered with sequential low-frequency (1 Hz) glutamate uncaging of individual spines of groups of targeted synapses (Oh et al., 2015). Such a paradigm involving a slow integration time window between stimulation would favor activation of calcineurin over that of CaMKII (Fujii et al., 2013), and hetLTD would dominate. In contrast, stimulation to elicit LTP that is confined to a single synapse might not reach the activation threshold of calcineurin while favoring the supralinear activation of CaMKII; without hetLTD, heterosynaptic LTP or metaplasticity for subsequent potentiation would be promoted. In our study, we stimulated groups of synapses quasi-synchronously by using glutamate photolysis, which resulted in the activation of both CaMKII and calcineurin, as supported by the sensitivity to inhibitors. Although the postsynaptic neuron was depolarized during glutamate photolysis, under such LTP triggering stimulation, a trend for a higher rise in calcium was observed for proximal spines located within 3 μm of the stimulated cluster than more distal spines (Figure S2). Locally boosted calcineurin activation by the spatially clustered stimulation, therefore, could have promoted the shrinkage of spines close to the stimulated spines, whereas a potentially more efficient spread of CaMKII signaling than those of calcineurin and/or a less strongly thresholded CaMKII activation at distal spines could have promoted their potentiation. Taken together, the expression of heterosynaptic plasticity at individual synapses is strongly biased by ongoing spatiotemporal patterns of synaptic activity. Here, we discussed the simplest possibility in an attempt to explain the observed simultaneous engagement of heterosynaptic potentiation and depression, based on the presumed differential activation thresholds and diffusional properties of calcineurin and CaMKII signaling. Nevertheless, interactions of synaptic scaffolding proteins and regulatory proteins with calcineurin and CaMKII at individual spines (Dudilot et al., 2020; Woolfrey and Dell'Acqua, 2015) are also likely to contribute to the landscape of heterosynaptic responses. A better understanding of the coordinated responses of signaling networks to patterned activity across neighboring synapses is warranted.

Here, somewhat unexpectedly, our observations suggest a requirement for NO in presynaptic hetLTD. Although NO has been traditionally studied in the context of presynaptic LTP (e.g., Arancio et al., 1996; Stanton et al., 2005; Ratnayaka et al., 2012; Padamsey et al., 2017), several studies have implicated NO in homosynaptic LTD (Zhuo et al., 1994; Reyes-Harde et al., 1999; Stanton et al., 2003; Gage et al., 1997). These studies collectively support a putative model for NO-dependent

(B) Induction of spike-timing-dependent LTP led to long-lasting increase in the EPSP slope of the stimulated pathway (homosynaptic) and decrease in the EPSP slope of the unstimulated pathway (heterosynaptic). Example traces illustrate the average EPSP waveform before LTP induction and during the last 5 min of the recording of the stimulated (black) and unstimulated (red) pathway.
(C) PPR before and after the induction of LTP. The stimulated pathway showed a significant decrease of the PPR (Wilcoxon rank-sum test). Conversely, the PPR increased in the unstimulated pathway.
(D) Bath application of 100 μM L-NAME abolished heterosynaptic LTD but did not affect the magnitude of homosynaptic LTP. Spread of LTP was unmasked in the unstimulated pathway.
(E) PPR changes of both stimulated and unstimulated pathways were abolished.
(F) Summary of EPSP slope changes.
(G) Summary of PPR changes (see also Figure S3). Statistical significance was assessed using Mann-Whitney U test. Data points are shown as mean ± SEM. See also Figure S5.

LTD at Schaffer-collateral-CA1 synapses that involves the activation of presynaptic internal Ca^{2+} stores by cyclic guanosine monophosphate (cGMP)-dependent protein kinase (PKG), a known downstream effector of NO signaling, along with ADP-ribosyl cyclase and ryanodine receptor, and also requires the concurrent activation of presynaptic CaMKII (Reyes-Harde et al., 1999; Stanton and Gage, 1996; Gage et al., 1997). The involvement of CaMKII in NO-mediated presynaptic depression is in line with our observation that inhibiting CaMKII abolished presynaptic hetLTD, although we did not discern the exact locus of CaMKII activation in our experiments. Whether or not postsynaptically generated NO at the stimulated spines diffuses short distances to neighboring presynaptic terminals remains to be confirmed, although the spatially confined heterosynaptic spread of presynaptic LTD (within 4 μm of stimulated cluster) is consistent with the direct involvement of short-lived NO as the extracellularly diffusible messenger.

Previously, a crucial role for NO in presynaptic LTP was demonstrated by the Emptage group (Padamsey et al., 2017). Specifically, NO synthesis driven by activation of L-type voltage-gated Ca^{2+} channels (L-VGCCs) coupled to presynaptic activity promoted an increase in Pr. The NO in the current study was unlikely to have originated by the same mechanism due to the fast inactivation of L-VGCCs at depolarized membrane potential used during the induction of cl-LTP. Such differences between the two studies suggest distinct functional roles for NO that depend on the context in which NO is generated. In the previous study, local photolysis of NO at single synapses on its own was not sufficient to induce LTP or LTD in the absence of presynaptic stimulation, whereas after dialyzing caged NO into the postsynaptic neuron, photolyzing a larger area could elicit LTD (Figure 5 Suppl. 2 in Padamsey et al., 2017). Therefore, induction of presynaptic hetLTD might require a critical amount of NO in conjunction with synaptic activity that is dependent on NMDARs but not L-VGCCs. In contrast, L-VGCC is needed for NO-dependent presynaptic potentiation.

Lastly, our study identified a slight trend for pre- and postsynapses to undergo correlated changes. Nevertheless, the correlation was not strong, and moreover, an apparent lack of correlation was noted when we inhibited calcineurin and observed presynaptic LTD in combination with postsynaptic LTP. The observed uncoupling of the direction of plasticity and dissociation of the pharmacological sensitivity of pre- and postsynaptic changes suggest that the pre- and the postsynaptic plasticity might represent in part independent parallel processes that follow unique dynamics. That the pre- and the postsynaptic sides of the same synapse do not necessarily change their strengths in the same direction may seem counterintuitive based on the overall matching of the pre- and the postsynaptic strengths at basal state across a synapse population (Tokuoka and Goda, 2008; Kay et al., 2011). Nevertheless, a similar disjunctional plasticity of presynaptic potentiation associated with postsynaptic depression was observed in cultured hippocampal neurons (Xu et al., 2013; Letellier et al., 2019) and between layer 4 neurons in visual cortical slices (Sáez and Friedlander, 2009). Such a mechanism could confer homeostatic stability to synapses, although the precise properties of disjunctional plasticity warrant further studies. Notably, the independent regulation of pre- and postsynaptic strengths has recently been explained by the difference in functional impact on the postsynaptic neuron (Costa et al., 2017, Tong et al., 2020). For future studies on the regulation of synaptic strength, it is therefore crucial to consider both, pre- and postsynaptic strengths for fully interpreting the consequences on neural network activity.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- RESOURCE AVAILABILITY
  - Lead contact
  - Materials availability
  - Data and code availability
- EXPERIMENTAL MODEL AND SUBJECT DETAILS
- METHOD DETAILS
  - Preparation of organotypic hippocampal slice culture
  - Preparation of acute hippocampal slices
  - 2-photon imaging
  - Optical quantal analysis
  - Glutamate photolysis
  - Induction of structural LTP
  - Analysis of spine structural changes
  - Spine calcium imaging
  - Electrophysiological recordings in acute slices
  - Pharmacology
- QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at https://doi.org/10.1016/j.celrep.2021.108693.

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AUTHOR CONTRIBUTIONS

R.T., Y.G., and N.J.E. conceived the study and designed the experiments. R.T. conducted and analyzed the experiments. T.E.C. performed and analyzed the calcium imaging experiments. The manuscript was drafted by R.T., and all authors edited and approved the final manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
Oregon Green 488 BAPTA-1 (hexapotassium salt)	ThermoFisher Scientific	Cat#O6806
Fluo-5F	ThermoFisher Scientific	Cat#F14221
AlexaFluor 488	ThermoFisher Scientific	Cat#A10436
AlexaFluor 594	ThermoFisher Scientific	Cat# A10438
MNI-glutamate	Tocris	Cat#1490; CAS: 295325-62-1
AP5	Abcam	Cat#ab120003; CAS: 79055-68-8
KN62	Tocris	Cat#1277; CAS: 127191-97-3
FK-506	Tocris	Cat#3631; CAS: 104987-11-3
L-NAME	Cayman Chemical	Cat#Cay80210; CAS: 51298-62-5
Picrotoxin	Sigma Aldrich	Cat#P1675; CAS: 124-87-8
Trolox	Sigma Aldrich	Cat#238813; CAS: 53188-07-1
Ascorbic acid	Sigma Aldrich	Cat#A92902; CAS: 50-81-7
Experimental models: organisms/strains		
Rat: WI Wistar rat (rattus norvegicus)	Harlan UK, Nihon SLC	RRID: RGD_13508588
Software and algorithms		
ImageJ	NIH	https://imagej.nih.gov/ij/
pClamp	Molecular Devices	RRID: SCR_011323
Zen 2009	Zeiss Microscopy	https://www.zeiss.com/corporate/int/home.html
Other		
Zeiss LSM780	Zeiss Microscopy	https://www.zeiss.com/corporate/int/home.html
Leica DMLFSA	Leica	https://www.leica-microsystems.com

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Yukiko Goda (yukiko.goda@riken.jp).

Materials availability

This study did not generate new unique reagents.

Data and code availability

The data generated in the current study and analysis code have not been deposited in a public repository but are available from corresponding authors upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Male Wistar rat pups (RRID: RGD_13508588, age: postnatal day 6-7 and 14-21) were used for making organotypic and acute hippocampal slices. Animals were supplied by Harlan UK and Nihon SLC. All animal experiments were either approved by the RIKEN Animal Experiments Committee and performed in accordance with the RIKEN rules and guidelines or were in accordance with the Animals (Scientific Procedures) Act, 1986 (UK).

METHOD DETAILS

Preparation of organotypic hippocampal slice culture

Organotypic hippocampal slices were prepared as previously reported (Padamsey et al., 2017). Hippocampi of postnatal day P6-7 Wistar rat pups (Harlan UK, Nihon SLC) were isolated and cut into 350 μm-thick transverse slices on a McIlwain tissue chopper (Mickle Laboratory Engineering Co. Ltd. and Cavey Laboratory Engineering Co. Ltd.). The brain was dissected in ice-cold Earle's Balanced Salt solution (EBSS) containing 35 mM glucose, 20-25 mM HEPES and adjusted to pH 7.35 with 5 mM NaOH. Slices were transferred onto cell culture inserts (0.4 μm pore size, Merk Millipore) and placed in a 6-well plate filled with 1 ml/well of culture media containing 50% Minimum Essential Medium (MEM, Thermo Fisher Scientific), 23% EBSS, 25% horse serum (Thermo Fisher Scientific), and 36 mM glucose. Some slices (used for experiments in Figures 2 and 3) were cultured in media that included 2% B-27 Supplement (Thermo Fisher Scientific). Slices were maintained at 37°C and 5% CO_2 and used for experiments at DIV10-15.

During experiments, slices were constantly perfused (1-2 ml/min) with artificial cerebrospinal fluid (aCSF) containing (in mM) 145 NaCl, 2.5 KCl, 16 $NaHCO_3$, 1.2 NaH_2PO_4, 11 glucose, 3 $CaCl_2$, and 2 $MgCl_2$. For experiments with intensive imaging, 1 mM Trolox (Sigma Aldrich) and 0.2 mM ascorbic acid were additionally included. The aCSF was bubbled with 95% O_2 and 5% CO_2 and heated to near-physiological temperature (33-35°C) using an in-line heater (Warner Instruments).

Preparation of acute hippocampal slices

Acute hippocampal slices were prepared from P14-21 male Wistar rats (Harlan UK). Rats were sacrificed and brains were extracted and immediately submerged in ice-cold dissection media saturated with 95% O_2 and 5% CO_2. Dissection media consisted of (in mM): 65 sucrose, 85 NaCl, 2.5 KCl, 25 $NaHCO_3$, 1.25 NaH_2PO_4, 10 glucose, 7 $MgCl_2$, and 0.2 $CaCl_2$. The cerebellum was manually removed by a coronal cut using a razor blade to create a flat surface, and the brain was glued onto a platform of the vibratome (Microm HM 650V, Thermo Scientific). Brains were further stabilized by positioning a block of 2% agar to contact the dorsal surface of the brain. 400 μm coronal slices were cut, and the hemispheres were separated and transferred into a recovery chamber containing standard aCSF (in mM: 120 NaCl, 2.5 KCl, 26 $NaHCO_3$, 1.2 NaH_2PO_4, 11 glucose, 1 $MgCl_2$, and 2 $CaCl_2$), which was bubbled with 95% O_2 and 5% CO_2. Slices were allowed to recover for 8 min at 37°C and 60 min at room temperature prior to use. Slices were maintained at room temperature up to 4-5 h.

For experiments, slices were transferred to the recording chamber and secured using a "harp" (Warner Instruments). The recording chamber was constantly perfused (2-3 ml/min) with aCSF, and 100 μM picrotoxin (Sigma Aldrich) was added to block inhibitory synaptic transmission. The aCSF was bubbled with 95% O_2 and 5% CO_2 and heated to near-physiological temperature (33-35°C) using a custom made in-line heater.

2-photon imaging

CA1 pyramidal cells were bolus-loaded with 0.5 mM AF594 (ThermoFisher Scientific) and 1 mM Oregon Green BAPTA-1 (OGB-1, ThermoFisher Scientific) diluted in standard internal solution (in mM: 135 K-Gluconate, 10 KCl, 10 HEPES, 1 $MgCl_2$, 2 Na_2ATP, and 0.4 Na_3GTP) for 45-60 s. OGB-1 was included for optical quantal analysis (see Optical quantal analysis below). The dyes were given at least 10 min and up to 45 min to allow for sufficient diffusion in proximal dendrites such that no obvious change in mean fluorescence intensity over time was observed (Padamsey et al., 2017). After identifying a segment of dendrite with linear geometry that was largely contained within a single optical plane and hence suited for imaging, presynaptic strength was measured via optical quantal analysis. Afterward, z stacks (0.5 μm steps) capturing ~35 μm of the dendrite centered on the previously measured spine were taken every 5 min. Images were acquired on a Zeiss LSM780 confocal laser scanning microscope with a 63x 1.0 NA objective (Plan-APOCHROMAT, Zeiss) using commercial software provided by Zeiss (Zen 2009, Version 6.0.0.303). Images were taken at 4x zoom, which led to a lateral pixel size of 65.9 nm. Both fluorescent dyes were simultaneously excited using an 800 nm 2-photon laser source (Coherent) and emission was separated using bandpass emission filters. At least three baseline images were taken before applying clustered stimulation to elicit LTP (cl-LTP). Images were then taken at 1, 5, 10, 20, 30, and 45 min after the cl-LTP induction. Experiments showing elevated Ca^{2+} levels or outgrowth of filopodia were discarded as these features correlated to subsequent signs of cell death, such as fragmentation of the dendrite. At the end of the experiment, a z stack was taken (2 μm steps, 1x zoom), which included the dendrite of interest, the cell body, and the position of the stimulation and puffing electrodes.

Optical quantal analysis

We performed optical quantal analysis as previously reported (Padamsey et al., 2019). A stimulation electrode (tungsten monopolar in glass pipette) was positioned within 5-10 μm of the dendrite of interest. xt-scans (line scans) covering large fractions of spines were taken at 500 Hz while synapses were stimulated with a paired pulse given at 70 ms inter-pulse interval. The imaging software was synchronized with Clampex software (Molecular Devices) via TTL pulses. Stimulation and the initiation of image acquisition were controlled by Clampex. Responsive spines were detected as time-locked OGB-1 Ca^{2+} responses (EPSCaTs). In order to prevent potential confounds of AP failures, which will inflate the estimate of release failures, the stimulation strength was increased until the observed Pr stabilized, such that the average failure rate remained constant over time (Emptage et al., 1999). For data acquisition, line scans were oriented to capture both the spine of interest and the dendrite, which allowed also the detection of local dendritic

spikes or back-propagating APs. Before the induction of cl-LTP, optical quantal analysis was limited to 20-25 trials (inter-trial interval: 15 s) to prevent photodynamic damage. Optical quantal analysis was repeated 30 min after the induction of LTP and 20-30 trials were taken. The stimulation strength was further increased at the end of the experiment and a paired-pulse at 5 ms inter-pulse interval was given to ensure that release events could still be detected. This also ruled out the possibility that an observed decrease in Pr is due to spatial drift of the stimulation electrode, a change in action potential initiation, or a decrease in the EPSCaT amplitude below detection threshold. Experiments were excluded if no release events could be detected. EPSCaTs were analyzed in ImageJ (Schindelin et al., 2012; Schneider et al., 2012) and a custom written Python script. Fluorescence signals were background-subtracted, averaged within the spatial dimension, and normalized to baseline (defined as the average fluorescence before the first stimulation pulse). Successful release events were counted manually. All results were cross-validated using an automated analysis script, which detected successful release as an increase in the average Ca^{2+} signal within 6 ms after stimulation that exceeded two standard deviations above the mean of the baseline. No qualitative difference was found between the two methods; however, the automated method was not robust against imaging artifacts or unstable baselines. To distinguish successful release events from dendritic or somatic spikes, the peak intensity was required to be at least 50% larger in the spine compared to the neighboring dendrite. Alternatively, the identification of EPSCaTs was aided by the signal onset that preceded dendritic/somatic spikes. Pr was calculated as the ratio between the number of successful release events to the first stimulation pulse and number of trials. The sampling error is given by the binomial theorem

$$\sigma = \sqrt{\frac{p(1-p)}{N}}$$

where p is the release probability and N the sample number. For an average sample number of N = 25, the maximum expected error is therefore ~0.1.

Glutamate photolysis

Caged glutamate was locally applied during photolysis through a glass pipette coupled to a picospritzer (5-10 psi, Parker Instrumentation). 10 mM MNI-glutamate (Tocris) dissolved in Tyrode's solution (in mM: 58.44 NaCl, 2.5 KCl, 20 HEPES, 30 glucose, 3 $CaCl_2$, 2 $MgCl_2$, pH-adjusted to 7.2 using 5 mM NaOH) was filtered (0.45 μm pore syringe filter, Merck Millipore) and loaded into a glass pipette (3-4 MΩ). The pipette was positioned ~10-20 μm from the dendrite of interest. A 2-photon laser source (720 nm) was used for focal photolysis of glutamate. Photolysis was controlled by a custom written script for the imaging software and was synchronized with the electrophysiology via Clampex. Photolysis consisted of 4 ms pulses, and the laser power was adjusted for each experiment to obtain Ca^{2+} transients similar in size compared to EPSCaTs measured in the same cell (laser power varied from 2.4-3.5 mW measured at the sample). For the induction of clustered LTP, laser power was reduced slightly (to ~80%) to prevent over-excitation, which would frequently cause excitotoxicity observed as sustained elevated Ca^{2+} in the dendrite. For quasi-synchronous photolysis, the inter-pulse interval was ~2.2 ms.

Induction of structural LTP

Clustered structural LTP (cl-LTP) was induced by quasi-synchronous glutamate photolysis at 5-7 spines paired with postsynaptic depolarization, 30× at 0.5 Hz. The postsynaptic neuron was re-patched with internal solution containing 100 μM OGB-1 and 50 μM AF594 to reduce the osmotic pressure of the dye. Photolysis was set up after the GΩ-seal was established by focusing on the focal plane containing at least 5 neighboring spine heads, and cl-LTP was initiated within 10 s of establishing whole-cell mode in order to prevent dialysis. Cells were held at 0 mV in voltage clamp mode. The average access conductance was Ga = 45.86 ± 2.56 nS. Cells were allowed to reseal after the induction by gently retracting the patch pipette. The success rate of re-patch and reseal was ~80%. The maximum distance and mean pairwise distance between stimulated spines did not vary across experimental conditions.

Analysis of spine structural changes

Spine size was analyzed in ImageJ. The integrated fluorescence of a spine contained in a rectangular regions of interest (ROI) was used as an estimate of spine size. Integrated fluorescence was measured on the z-image slice that resulted in the maximal value. The background fluorescence equivalent to the size of the ROI was subtracted. The re-patch during clustered LTP induction caused a substantial global reduction in the fluorescence intensity. Fluorescence signals were therefore normalized to the mean fluorescence intensity of the underlying dendrite, which was assumed to remain structurally stable (Nimchinsky et al., 2004). At least three dendritic segments were averaged to calculate the mean fluorescence intensity of the dendrite. Spine size changes were calculated as fractional changes with respect to the average size prior to cl-LTP induction. Final spine size changes (at the end of the experiment) was defined as the average change observed at 30 and 45 min.

All visible spines along the dendrite of interest were analyzed. Spines that were only partially visible or those located within 2 μm of the photolysis spot were excluded from the analysis. Spines that showed substantial fluctuations (> 25%) during the baseline were also excluded, although these were rare (< 5% of all spines analyzed). For subgroup analysis of spines, such as distance bins, spine structural changes were averaged within experiments before averaging across experiments.

Spine calcium imaging

Hippocampal organotypic slices were continually perfused with aCSF as described above, with the addition of 4 mM MNI-glutamate. Individual CA1 pyramidal neurons were patched with intracellular solution containing (in mM) 100 K-gluconate, 17 KCl, 5 NaCl, 5 $MgCl_2$, 10 HEPES, 0.5 EGTA, 4 K_2ATP, and 0.5 Na_3GTP (pH 7.3, osmolarity adjusted to 280 mOsm). On the day of the experiment 200 μM Fluo-5F and 30 μM Alexa 594 were added to the intracellular solution. After allowing cells to fill for 15 min, dendritic segments were chosen for imaging. Dendrites were imaged at 2 Hz using 800 nm laser light during which dendrites received one of three stimuli: depolarization alone, MNI-glutamate photolysis alone, or a combination of both depolarization and MNI-glutamate photolysis. For depolarization, the cell was held for 60 s at 0 mV. For glutamate photolysis, 30× 4 ms pulses of 720 nm laser light were delivered 0.5 μm from spine heads at 0.5 Hz. For the depolarization alone condition, sham homosynapses were selected prior to imaging. Images were analyzed in ImageJ. Individual spines were overlaid with ROIs, background subtracted, and Fluo-5F intensity was measured for every time point throughout the experiment. Spine calcium levels were then normalized to the pre-stimulation baseline. Peak calcium was defined as the average of the first 10 images immediately following stimulation. Plateau calcium was defined as the average of the stable calcium transient that followed the peak (70 images, beginning 20 s after stimulation onset). Cells were discarded if calcium levels did not return to pre-stimulation levels.

Electrophysiological recordings in acute slices

CA1 pyramidal cells located 50-100 μm below the slice surface were targeted for patch clamp recordings. For visualization of the dendritic arbour, 200 μM AlexaFluor 488 (AF488) was included in the internal solution and cells were bolus-loaded for 5 min and patched-off by gently retracting the patch pipette. The fluorescence signal was captured on a Leica DMLFSA confocal microscope with a 63x water-immersion objective (NA = 0.9, HCX APO L 63x/0.9 W U-V-I, Leica) using the Leica Confocal Software (Version 2.61). Neurons were stimulated using two monopolar tungsten electrodes placed directly into the tissue. In order to stimulate sets of synapses in close spatial proximity along individual segments of dendrite, stimulation electrodes were positioned within 10-20 μm of a visually-identified segment of dendrite, 10-40 μm apart from each other. The position of the stimulation electrode could either be visualized via autofluorescence or as shadows in cases where tissue autofluorescence was strong. After positioning, a 10 min wait was included for the stimulation electrodes to reach thermal equilibrium in order to minimize spatial drift. Cells were subsequently re-patched with internal solution containing 200 μM AF488. The probability of successful re-patch was > 90%. Re-patch of the correct cell was confirmed by fluorescence imaging.

A cross-facilitation test was used to ensure stimulation of two independent pathways. The two pathways were stimulated sequentially with 70 ms inter-pulse interval followed by stimulation in the reverse order, repeated 10 times. A lack of detection of short-term facilitation of synaptic responses to the second pulse was used as a criterion for independence of the two pathways. The magnitude of overlap was estimated by comparing the fractional difference in EPSP slope of a given pathway when stimulated as first or second pulse. Significant overlap was rarely observed, which was also supported by the lack of post-tetanic potentiation in the unstimulated pathway.

EPSPs were evoked at 0.06 Hz by 100 μs current pulses (20-80 μA), alternated between the stimulation electrodes. PPR was measured every fifth sweep. A stable baseline was recorded for a maximum of 8 min in order to reduce potential confounds due to dialysis of cytosolic contents. Experiments showing consistent run-up or run-down were discarded. LTP was induced in one pathway (randomly chosen) by pairing stimulation with strong postsynaptic depolarization via current injection, 60× at 5 Hz, as previously reported (Padamsey et al., 2017). The unstimulated pathway was kept silent during the induction. EPSPs were recorded for at least 30 min after the induction and PPRs were sampled throughout the recording period.

Pharmacology

CaMKII was blocked by bath application of 10 μM of the non-competitive inhibitor KN62 (Tocris). Due to its low solubility in water, KN62 was first dissolved in DMSO and subsequently diluted in aCSF heated to 37°C. Calcineurin was blocked by bath application of 2 μM FK506 (Tocris). Both, KN62 and FK506 were maintained in the aCSF throughout the experiment. NO signaling was blocked by bath application of 100 μM L-NAME (Cayman Chemical), a nitric oxide synthase inhibitor. L-NAME was pre-incubated for at least 1 h before the experiment to abolish tonic levels of NO. NMDA receptor signaling was blocked by local perfusion of 500 μM AP5 (Abcam) diluted in Tyrode's solution (additionally containing 10 mM MNI-glutamate for glutamate photolysis).

QUANTIFICATION AND STATISTICAL ANALYSIS

Non-parametric tests were used for all statistical comparisons. Mann-Whitney U test was used for comparisons of independent means; Wilcoxon signed-rank test was used for dependent means. For multiple comparisons, Kruskal-Wallis H test was used followed by post hoc Dunn's test for pairwise comparisons. We report the coefficient of determination (r^2) for linear regression and further evaluated linear correlations using the Pearson correlation coefficient (r). Sample sizes (n) are indicated in the corresponding sections of the main text and refer to the number of experiments (note that each slice was used for a single experiment only). All measurements are stated as mean ± standard error of the mean (SEM). Significance is denoted as follows: *$p < 0.05$, **$p < 0.01$.